Week14HW

June 15, 2025

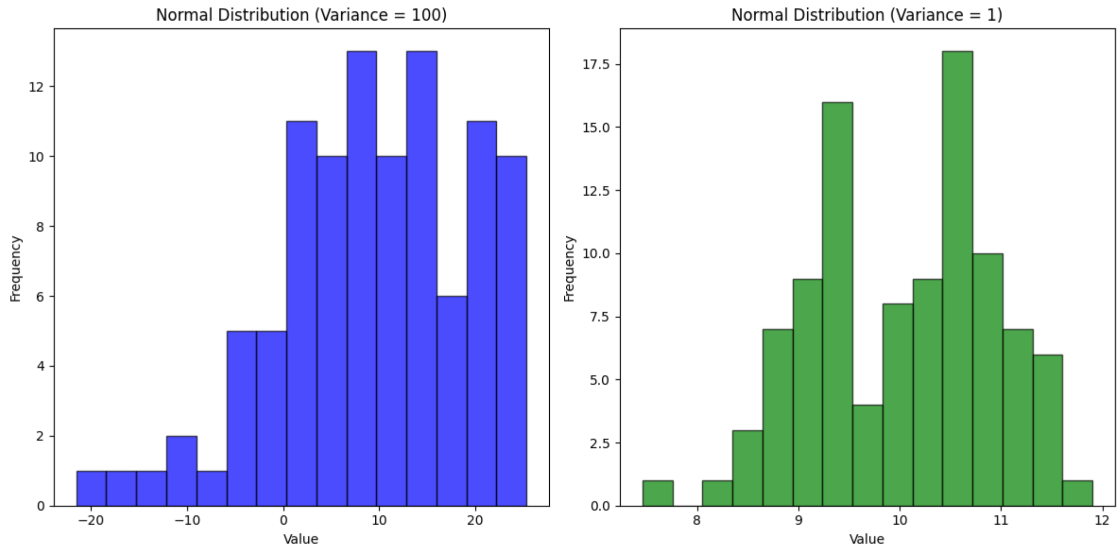
1 Week 14 Assignment

1.1 Problem 1

In the cell below, create a Normal distribution with mean 10 and variance 100. Draw 100 samples from the distribution and plot a histogram of the results. Repeat this experiement with the same mean but the variance set to 1. What differences do you notice?

[1]:
```
# Imports required
import numpy as np
import matplotlib.pyplot as plt

# Set the mean and variances
mean = 10
variance1 = 100
variance2 = 1

# Draw 100 samples from each distribution
samples_var1 = np.random.normal(mean, np.sqrt(variance1), 100)
samples_var2 = np.random.normal(mean, np.sqrt(variance2), 100)

# Plot histograms
plt.figure(figsize=(12, 6))

# Histogram for variance 100
plt.subplot(1, 2, 1)
plt.hist(samples_var1, bins=15, color='blue', alpha=0.7, edgecolor='black')
plt.title("Normal Distribution (Variance = 100)")
plt.xlabel("Value")
plt.ylabel("Frequency")

# Histogram for variance 1
plt.subplot(1, 2, 2)
plt.hist(samples_var2, bins=15, color='green', alpha=0.7, edgecolor='black')
plt.title("Normal Distribution (Variance = 1)")
plt.xlabel("Value")
plt.ylabel("Frequency")
```

```
# Show the plots
plt.tight_layout()
plt.show()


# What differences do you notice?
# I noticed with the distribution 100 it offeres a wider range while the normal␣
↪distribution of 1 it's more of a narrow distirbution.
#
```

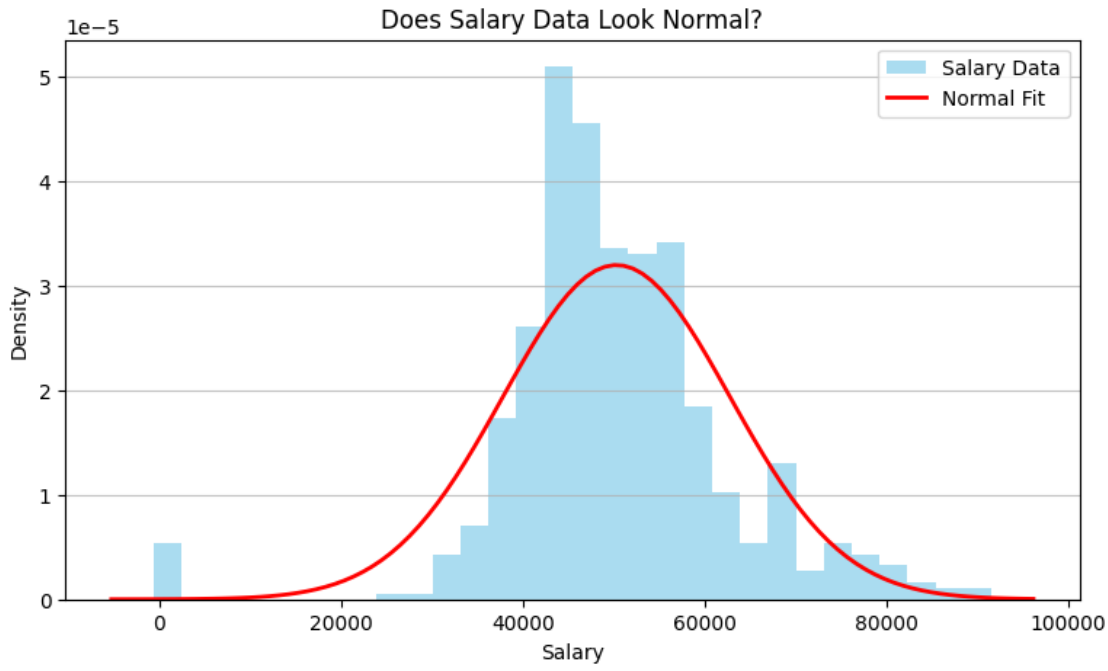


1.2 Problem 2

In the cell below, compare the salary data column from the `College_Rankings` data to a normal distribution. Does it look like a good fit?

[4]:
```
# imports needed
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import scipy.stats as stats

# Load data and clean 'Salary' column
try:
    df = pd.read_csv('College_Rankings.csv')
    salary_data = df['Salary'].dropna() # Corrected column name to 'Salary'
except FileNotFoundError:
    print("Error: College_Rankings.csv not found. Place it in the same␣
↪directory.")
```

```
    exit()

if len(salary_data) == 0:
    print("No salary data to analyze.")
else:
    # Plot histogram with normal curve
    plt.figure(figsize=(9, 5))
    plt.hist(salary_data, bins=30, density=True, alpha=0.7, color='skyblue',␣
 ↪label='Salary Data')
    mu, std = stats.norm.fit(salary_data)
    xmin, xmax = plt.xlim()
    x = np.linspace(xmin, xmax, 100)
    p = stats.norm.pdf(x, mu, std)
    plt.plot(x, p, 'red', linewidth=2, label='Normal Fit')
    plt.title('Does Salary Data Look Normal?')
    plt.xlabel('Salary')
    plt.ylabel('Density')
    plt.legend()
    plt.grid(axis='y', alpha=0.75)
    plt.show() # Use plt.show() if you want to display the plot immediately in␣
 ↪your environment

    # Print statement Q&A
    print("Does it look like a good fit?")
    print("I would say that the for the most part it looks like a good fit.")
```



```
Does it look like a good fit?
I would say that the for the most part it looks like a good fit.
```

1.3 Problem 3

In the cell below, repeat the regression analysis from the module workbook with the full HWAS data (not just the adults). What differences do you notice?

[ ]:

```

```